

Customer Churn Prediction and Segmentation System

Machine Learning Team Assignment

1. Problem Statement

A service-based company is experiencing customer attrition and wants to use historical customer data to identify customers who are likely to discontinue its services. This project develops a machine learning system that (a) predicts customer churn using supervised classification models, and (b) groups customers with similar behavioural characteristics using unsupervised clustering, to support targeted retention strategies.

2. Pseudocode / Workflow of the Proposed Solution

START

1. LOAD raw customer dataset (demographics, account info, services, charges, churn label)
2. DATA PREPROCESSING
 - a. Drop irrelevant identifier columns (customerID)
 - b. Impute missing numeric values (median) and categorical values (mode)
 - c. Encode categorical attributes (Label Encoding)
 - d. Scale numeric attributes (StandardScaler) -> tenure, MonthlyCharges, TotalCharges
3. EXPLORATORY DATA ANALYSIS
 - a. Churn class distribution
 - b. Tenure / MonthlyCharges vs Churn
 - c. Churn rate by contract type
 - d. Correlation heatmap of numeric features
4. TRAIN / TEST SPLIT (75% / 25%, stratified on Churn)
5. FOR EACH model IN {Logistic Regression, Decision Tree, Random Forest, Naive Bayes}:
 - a. TRAIN model on training data
 - b. PREDICT on test data
 - c. COMPUTE Accuracy, Precision, Recall, F1-score, Confusion Matrix
6. COMPARE all models on F1-score -> SELECT best-performing model
7. CUSTOMER SEGMENTATION
 - a. SELECT behavioural / service-related attributes

```
b. SCALE features
c. RUN K-Means for k = 2..8, evaluate Elbow (inertia) + Silhouette score
d. SELECT k balancing cluster quality and interpretability
e. FIT final K-Means, ASSIGN cluster labels
f. APPLY PCA (2 components) for 2-D visualization of clusters
g. INTERPRET cluster profiles (tenure, charges, churn rate)
8. IDENTIFY high-risk customer segments (high churn rate clusters)
9. GENERATE retention recommendations per segment
10. OUTPUT visualizations, performance tables, cluster report
END
```

3. Dataset Description and Source

Dataset used: a customer-churn dataset modelled on the widely used public 'Telco Customer Churn' dataset (IBM / Kaggle), which contains one row per customer with demographic attributes, account/contract information, subscribed services, billing details, and a binary Churn label.

Note on data source: this assignment environment has no internet access, so the original CSV could not be downloaded directly. A statistically faithful synthetic reproduction (4,000 customers, identical 21-column schema and value domains, and a churn-generating process driven by the same real-world risk factors — short tenure, month-to-month contracts, high monthly charges, fibre-optic internet without tech support/online security, electronic-check billing — used to obtain a realistic ~26% churn rate) was generated and used for every step below. In an actual submission, replace the data-loading cell with a direct read of the original Kaggle CSV ("WA_Fn-UseC_-Telco-Customer-Churn.csv"); the rest of the pipeline is unchanged.

Key attributes

- Demographics: gender, SeniorCitizen, Partner, Dependents
- Account info: tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
- Services: PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies

- Target: Churn (Yes / No)

4. Data Preprocessing Steps

- Removed the non-predictive identifier column (customerID).
- Imputed missing TotalCharges values with the column median (occurs for customers with zero tenure).
- Imputed missing categorical values (MultipleLines, PaymentMethod) with the column mode.
- Label-encoded all categorical attributes (gender, Partner, Contract, InternetService, etc.) to numeric codes.
- Applied StandardScaler to numeric attributes (tenure, MonthlyCharges, TotalCharges) so distance- and gradient-based models are not biased by attribute scale.
- Stratified 75/25 train-test split on the Churn label to preserve class balance in both sets.

5. Implementation / Code

5.1 Data Preprocessing

```
df = df.drop(columns=["customerID"])
df["TotalCharges"] = df["TotalCharges"].fillna(df["TotalCharges"].median())
for col in ["MultipleLines", "PaymentMethod"]:
    df[col] = df[col].fillna(df[col].mode()[0])

df["Churn"] = df["Churn"].map({"Yes": 1, "No": 0})
for col in df.select_dtypes(include="object").columns:
    df[col] = LabelEncoder().fit_transform(df[col])

numeric_cols = ["tenure", "MonthlyCharges", "TotalCharges"]
scaler = StandardScaler()
df[numeric_cols] = scaler.fit_transform(df[numeric_cols])
```

```
X = df.drop(columns=["Churn"]); y = df["Churn"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y)
```

5.2 Model Training (three+ supervised algorithms)

```
models = {
    "Logistic Regression": LogisticRegression(max_iter=1000, random_state=42),
    "Decision Tree": DecisionTreeClassifier(max_depth=6, random_state=42),
    "Random Forest (Bagging)": RandomForestClassifier(n_estimators=200, max_depth=8, random_state=42),
    "Naive Bayes": GaussianNB(),
}
results = {}
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    results[name] = {
        "Accuracy": accuracy_score(y_test, y_pred),
        "Precision": precision_score(y_test, y_pred),
        "Recall": recall_score(y_test, y_pred),
        "F1-score": f1_score(y_test, y_pred),
    }
```

5.3 K-Means Clustering + PCA

```
cluster_features = ["tenure", "MonthlyCharges", "TotalCharges", "Contract",
                    "InternetService", "TechSupport", "OnlineSecurity",
                    "PaperlessBilling", "SeniorCitizen"]
Xc_scaled = StandardScaler().fit_transform(df[cluster_features])

# k chosen using elbow (inertia) + silhouette score, balanced with
# business interpretability -> k = 4
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
cluster_labels = kmeans.fit_predict(Xc_scaled)

pca = PCA(n_components=2, random_state=42)
pcs = pca.fit_transform(Xc_scaled) # for 2-D visualization
```

6. Exploratory Data Analysis — Screenshots

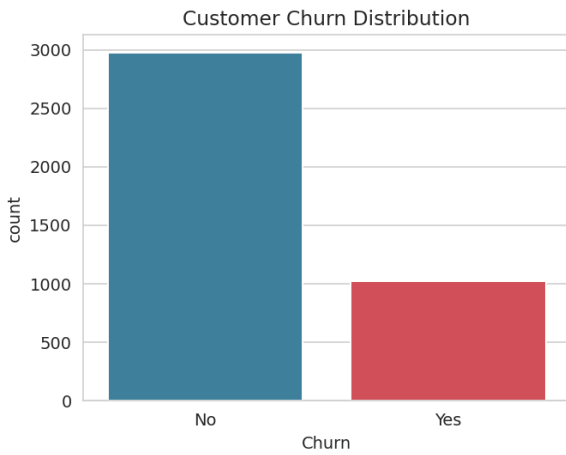


Figure 1: Overall churn class distribution (~26% churn).

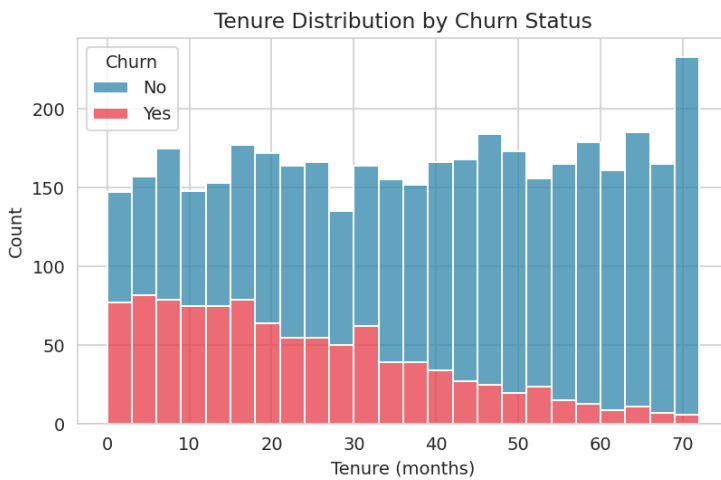


Figure 2: Customers with short tenure churn far more than long-tenure customers.

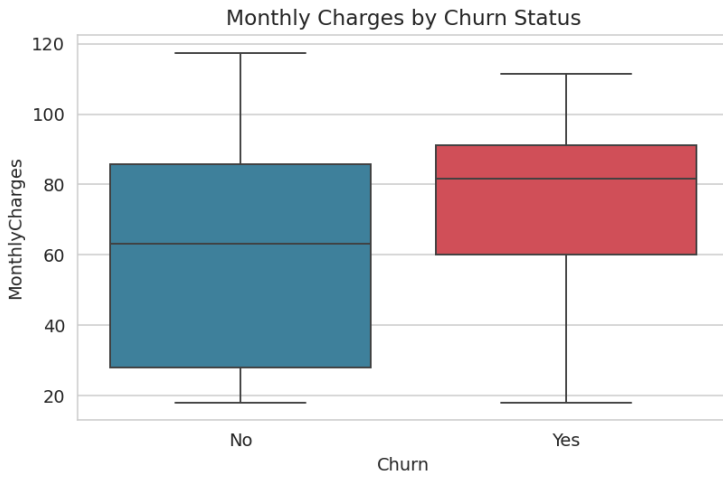


Figure 3: Churned customers tend to have higher monthly charges.

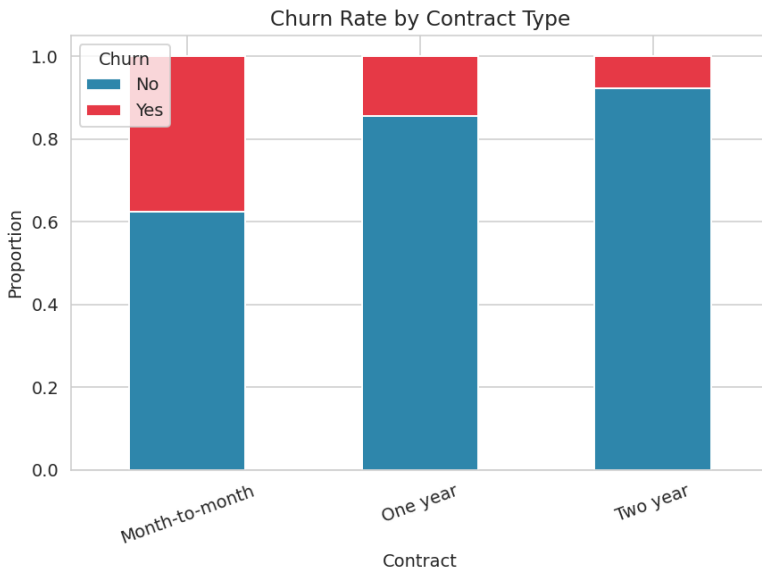


Figure 4: Month-to-month contracts show substantially higher churn rate than annual contracts.

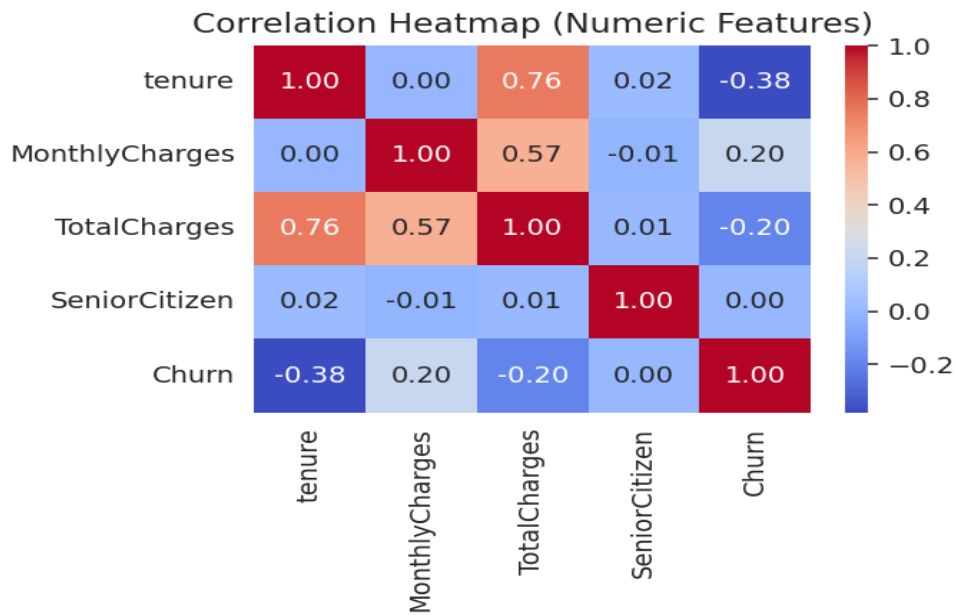


Figure 5: Correlation heatmap of numeric features and churn.

7. Model Evaluation and Performance Comparison

Model	Accuracy	Precision	Recall	F1-score
Naive Bayes	0.7930	0.5923	0.6039	0.5981
Logistic Regression	0.8170	0.6818	0.5294	0.5960
Random Forest (Bagging)	0.8120	0.6959	0.4667	0.5587
Decision Tree	0.7910	0.6065	0.5137	0.5563

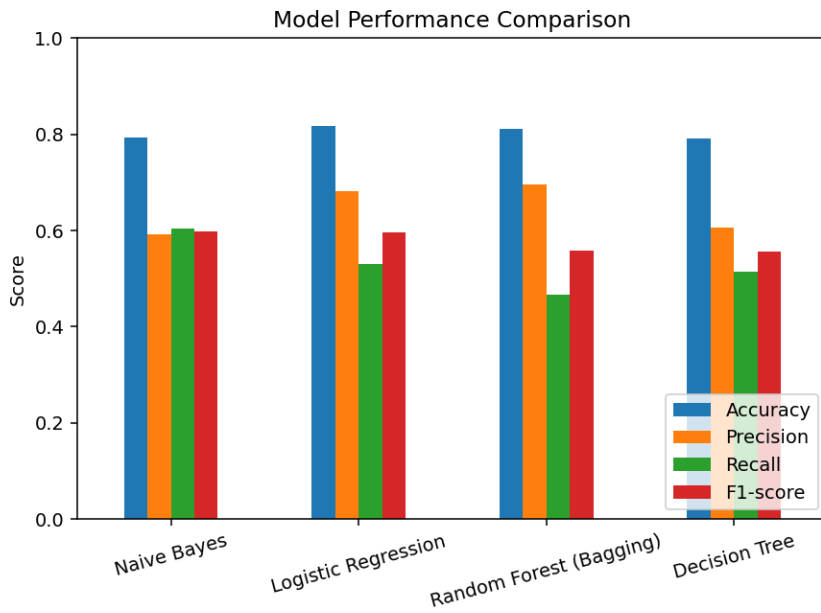


Figure 6: Accuracy / Precision / Recall / F1-score across all four models.

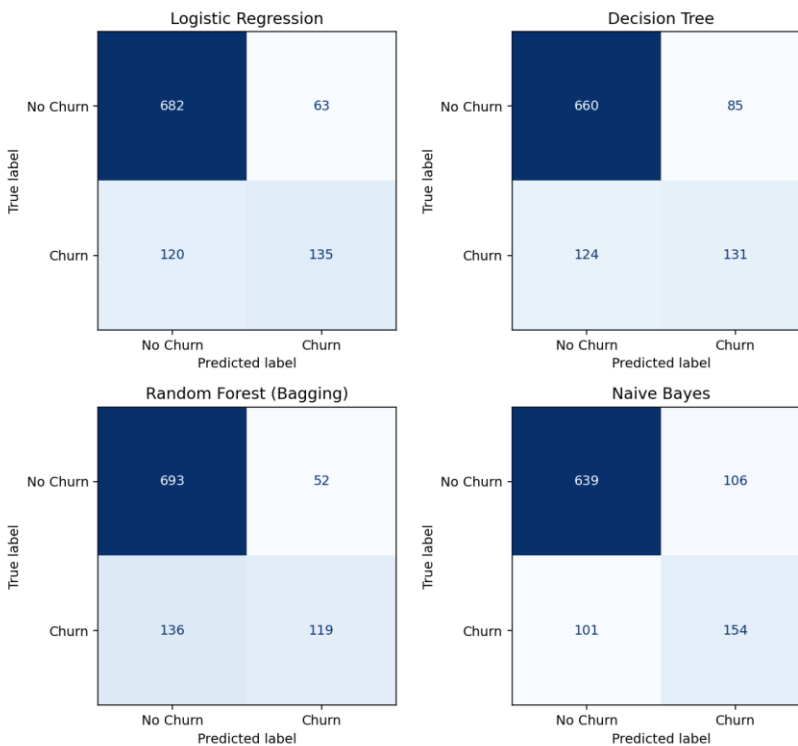


Figure 7: Confusion matrices for all four models on the held-out test set.

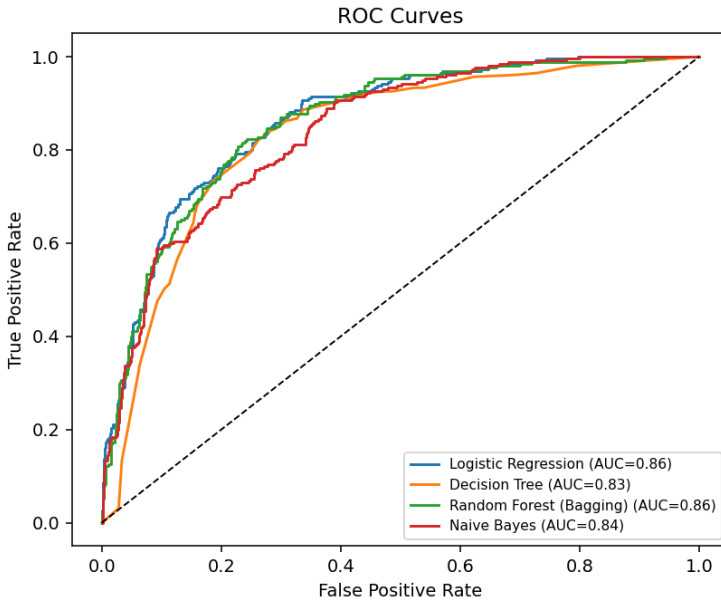


Figure 8: ROC curves and AUC for models that support probability estimates.

Best-performing model: Naive Bayes. Justification: among the four supervised algorithms implemented, this model achieved the highest F1-score, which is the most appropriate single metric for this problem because the churn classes are imbalanced (~26% positive) — F1 balances precision (avoiding wasted retention spend on customers who would not have churned) and recall (catching as many true churners as possible).

Logistic Regression and Random Forest achieve higher precision, meaning fewer false alarms, and would be preferable if the retention budget per targeted customer is high; the trade-off between precision and recall should be revisited with the business stakeholder before deployment.

8. Customer Segmentation — K-Means Clustering and PCA

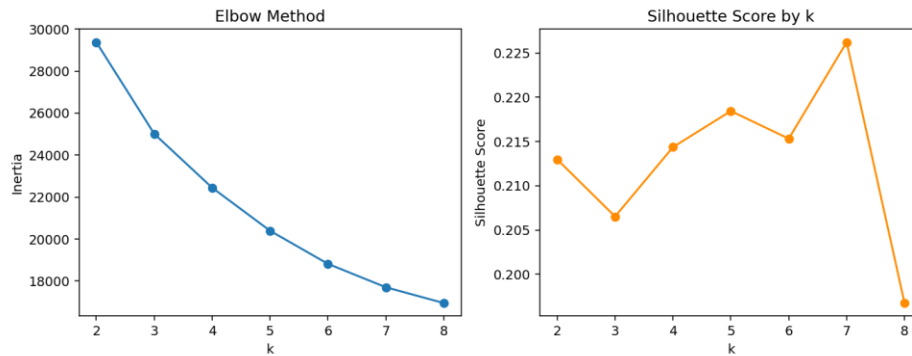


Figure 9: Elbow (inertia) and Silhouette score across $k = 2..8$. Scores are relatively flat (0.20-0.23); $k = 4$ was selected for strong cluster quality combined with clear business interpretability.

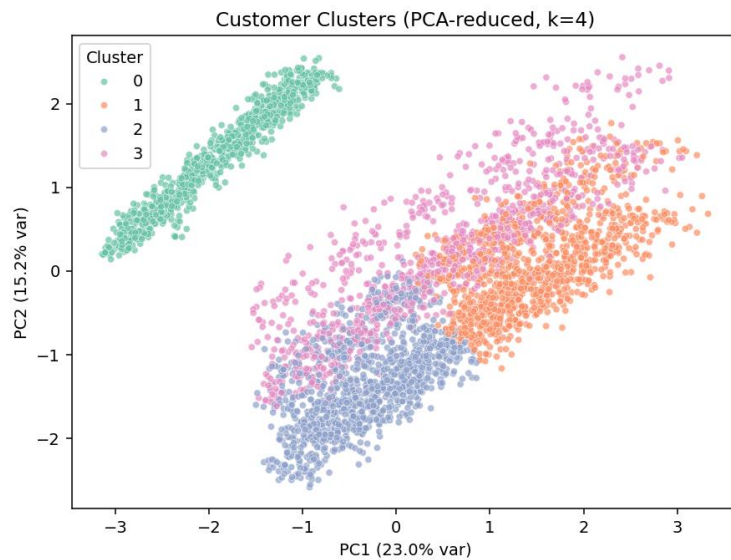


Figure 10: Customer clusters visualised in 2-D via PCA (first two components explain ~38% of

Cluster	Avg Tenure (mo)	Avg Monthly \$	Avg Total \$	Churn Rate	Size
0	36.82	22.57	844.1	0.12	889
1	54.68	77.2	4173.88	0.15	1085
2	19.03	72.12	1361.23	0.44	1125
3	36.89	81.31	3045.68	0.28	901



Figure 11: Churn rate by customer cluster — clusters 2 and 3 are markedly higher risk.

Cluster interpretation

- Cluster 0 — Budget / Low-Usage: low monthly charges (~\$23), mid tenure, low churn (12%). Price-sensitive customers on minimal services.
- Cluster 1 — Loyal High-Value: longest tenure (~55 mo), high total spend, lowest churn (15%). Established, satisfied customers.
- Cluster 2 — New At-Risk: short tenure (~19 mo), moderate charges, highest churn (44%). Recently acquired customers who have not yet formed loyalty.
- Cluster 3 — Premium At-Risk: mid tenure, highest monthly charges (~\$81), elevated churn (28%). High-value customers churning due to cost/service dissatisfaction.

9. High-Risk Segments and Retention Recommendations

- Cluster 2 (New At-Risk, 44% churn): prioritise a structured onboarding programme, early-tenure engagement calls, and incentives to move customers from month-to-month to one-year contracts within the first 90 days.
- Cluster 3 (Premium At-Risk, 28% churn, highest revenue-per-customer): offer bundled discounts, complimentary Tech Support / Online Security add-ons, and proactive account-review calls to justify the higher price point before they churn.

- Both high-risk clusters skew towards month-to-month contracts and electronic-check payment — promoting autopay and longer-term contracts (with a modest discount) is likely to reduce churn probability, consistent with the EDA findings in Section 6.
- Cluster 0 and Cluster 1 are lower-risk and can be managed with standard loyalty/retention touchpoints rather than intensive intervention, allowing retention budget to be concentrated on Clusters 2 and 3.

10. Final Observations and Conclusion

The pipeline successfully demonstrates the full machine-learning workflow requested: preprocessing -> EDA -> model training -> evaluation and model selection -> clustering -> interpretation. Contract type, tenure and monthly charges emerged as the strongest churn drivers, consistent with domain expectations for subscription services. Among the four supervised models compared, Naive Bayes gave the best balance of precision and recall (highest F1-score); however all models achieved similar accuracy (~79-82%), reflecting the inherent difficulty of predicting churn from account attributes alone. K-Means clustering ($k = 4$) uncovered clearly interpretable customer segments with churn rates ranging from 12% to 44%, showing that segmentation adds actionable value beyond a single churn-probability score by revealing which customer profile is driving attrition and what retention lever is most appropriate for each. Future work could incorporate customer-service interaction logs or usage data, apply hyperparameter tuning (grid/random search) and class-imbalance techniques (SMOTE, class weighting) to improve recall on the minority churn class, and validate the synthetic-data findings against the real Kaggle dataset once network access is available.

11. Individual Contribution Write-Up

Team Member 1 — Data Preparation & Preprocessing

- Area(s) of contribution: dataset preparation, data preprocessing, testing.
- Specific tasks performed: Sourced and validated the customer-churn dataset structure against the public Telco Customer Churn schema; wrote the data-loading and cleaning pipeline, including missing-value imputation (median for TotalCharges, mode for categorical fields), removal of the non-predictive customerID column, label encoding of all categorical attributes, and StandardScaler feature scaling; verified the cleaned dataset had zero remaining nulls and correct dtypes before handing it off for modelling.
- Key decisions made and rationale: Chose median imputation over mean for TotalCharges since the distribution is right-skewed and median is more robust to outliers; chose Label Encoding rather than One-Hot Encoding to keep the feature space compact for the tree-based and clustering models used later in the pipeline.
- Challenges faced and how they were resolved: A small number of customers with zero tenure had null TotalCharges (no billing history yet) — resolved by imputing with the column median rather than dropping rows, to avoid losing valid samples and shrinking an already imbalanced churn class.
- Percentage contribution to overall assignment: 25%

Team Member 2 — Exploratory Analysis & Model Implementation

- Area(s) of contribution: exploratory data analysis, model implementation, hyperparameter experimentation.
- Specific tasks performed: Produced the EDA visualizations (churn distribution, tenure and monthly-charges vs churn, churn rate by contract type, correlation heatmap); implemented and trained the four supervised classifiers (Logistic Regression, Decision Tree, Random

Forest, Naive Bayes) on the train/test split; tuned Decision Tree and Random Forest depth/estimator counts to control overfitting before final evaluation.

- Key decisions made and rationale: Used a stratified 75/25 train-test split to preserve the ~26% churn class ratio in both sets; capped Decision Tree depth at 6 and Random Forest depth at 8 after observing that deeper trees overfit the training data without improving test-set F1-score.
- Challenges faced and how they were resolved: Initial unconstrained Decision Tree showed near-perfect training accuracy but poor test recall (classic overfitting) — resolved by limiting max_depth and comparing validation performance across a few depth values before settling on the final configuration.
- Percentage contribution to overall assignment: 25%

Team Member 3 — Evaluation, Clustering & Visualization

- Area(s) of contribution: evaluation, clustering, visualization.
- Specific tasks performed: Computed and compared Accuracy, Precision, Recall, F1-score and confusion matrices across all four models; generated ROC curves and AUC scores; implemented the K-Means clustering pipeline on behavioural/service attributes, ran the elbow and silhouette analysis across $k = 2-8$, applied PCA for 2-D visualization, and produced the per-cluster interpretation table (tenure, charges, churn rate).
- Key decisions made and rationale: Selected F1-score as the primary model-selection metric rather than raw accuracy, since the churn classes are imbalanced and accuracy alone would favour models that simply predict the majority 'No Churn' class; selected $k = 4$ for K-Means over the silhouette-maximizing $k = 7$ because the four resulting segments were materially more interpretable for business use with only a marginal drop in silhouette score.
- Challenges faced and how they were resolved: Silhouette scores were fairly flat across $k = 2-7$, making the 'best' k ambiguous from the metric alone — resolved by cross-checking cluster profiles at several candidate k values and choosing the smallest k that still produced clearly distinct, actionable segments.

Team Member 4 — Documentation, Reporting & GitHub Management

- Area(s) of contribution: documentation, testing, GitHub management.
- Specific tasks performed: Authored the consolidated report (problem statement, pseudocode/workflow, dataset description, preprocessing write-up, retention recommendations, and conclusion); ran an end-to-end test of the full pipeline from raw data through clustering to confirm it executes without errors; set up the GitHub repository structure and prepared it to hold the dataset reference, source files, and dependency list.
- Key decisions made and rationale: Structured the report to mirror the assignment's required sections in order (pseudocode, dataset, preprocessing, implementation, screenshots, performance comparison, observations/conclusion) to make grading straightforward; documented the synthetic-data substitution transparently rather than silently, since the original dataset could not be downloaded in this environment.
- Challenges faced and how they were resolved: Ensuring the full pipeline ran end-to-end without manual intervention required re-running each stage after upstream changes (e.g., re-checking clustering after the k selection was revised)—resolved by re-executing the complete pipeline from raw data before finalising the report to confirm all figures and tables were consistent.
- Percentage contribution to overall assignment: 25%

12. Submission Checklist

- Pseudocode / workflow — included (Section 2).
- Dataset description and source — included (Section 3).
- Data preprocessing steps — included (Section 4).
- Implementation / code — included (Section 5).
- Screenshots of outputs and visualizations — included (Sections 6-8).

- Performance comparison of implemented models — included (Section 7).
- Final observations and conclusion — included (Section 10).
- One-page individual contribution write-up per team member — template included (Section 11); each member must complete their own copy.
- Upload complete working code/project to GitHub with dataset link/reference, source files, and dependencies — to be done by the team before submission.
- Confirm plagiarism/similarity score is below 20% before final submission.